

Full Marks 100

- $$\frac{1 \times 4 \times 9 + 2 \times 8 \times 12 + 3 \times 12 \times 27 + \dots + 2001 \times 8004 \times 18009}{1 \times 2 \times 3 + 2 \times 4 \times 6 + 3 \times 6 \times 9 + \dots + 2001 \times 4002 \times 6003}$$

- [illegible]

- A B C D E

$$\begin{array}{r} x\ 4 \\ \hline E\ D\ C\ B\ A \end{array}$$

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11. In a mathematics Olympiad exam there are 20 questions to be answered. For each correct answer 5 marks, for each wrong answer -2 and for each unanswered question 0 marks were awarded. A boy got 57 marks in that exam. How many questions did he answer correctly?

12. A and B were talking about their ages:

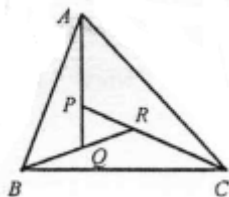
A: I am certainly not over 40

B: you are atleast 5 years older than me and I am 38 years old now.

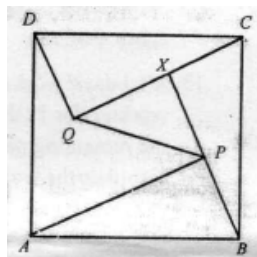
A: you are atleast 39.

It is known that all the statements made by A and B are false. How old are A and B?

13. A's 3 day's work is equal to B's 4 day's work. A and B together started doing a work. After working 12 days, A discontinued working due to illness. B continued and completed the remaining part of the work in 8 more days. In how many days could A and B each completed the work while working alone?
14. In the diagram AQ, BR and CP are drawn from the vertices A, B and C of triangle ABC. Points P, Q and R lie on AQ, BR, and CP respectively. If $AP = 3PD$, $BQ = 3QR$, $CR = 3PR$, then determine the ratio between the areas of triangle PQR and triangle ABC.



15. If 6588, 4516 and 9637 are divided by a fourth number the remainder in each case is same. What can be the greatest value of the fourth number?
16. Find the smallest number which when divided by 3, 5 and 7 respectively leaves remainder 1, 2 and 3 respectively.
17. A number has as digits 2000 ones and 2000 twos and all other digits are zeros. Can this number be a perfect square?
18. Find the remainder when $(1835)^{1910} + (1986)^{2061}$ is divided by 7.
19. In the given below diagram, ABCD is a square and $\angle APB$ is a right angle. If $CQ \parallel AP$, $DQ \parallel BP$ and X is a point on CQ, such that P lies on BX, then prove that the measure of $\angle PQX = 45^\circ$.



20. A merchant gets 17% profit even after giving 10% discount on the marked price on a certain item. By what percent should the marked price be increased, so that he will get 20% profit.
